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EDUCATION

University of California San Diego (UCSD) 2023 - June 2027 (expected)
B.S. Mechanical Engineering (MAE), Minor in Cognitive Science Design Cumulative GPA: 3.37

WORK EXPERIENCE

Rocket Lab USA September 2026 - Present
Mechanical Engineering Intern - Wiring and Harness Design

- Conducting **mechanical testing** of fir tree links to ensure secure harness mounting from **pull force and vibrations** on panels. Collaborating with mechanical team in hopes of integrating the component on all programs to minimize outgassing.
- Fabricating space vehicle **mockups** to conduct fit-check testing and harness integration. **3D printing** and **machining** metal connector mates for harnesses to latch and test lengths, hoping to build standardized modules to be reused on future programs.
- Drew RTD and heater **harness** wiring diagram in **Capital Essentials**, defined 50+ inline connections in space vehicle master schematic. Executed **CAD electrical routing** in **Siemens Nx** and generated 6 **formboard technical drawings** to direct physical manufacturing. Collaborating with thermal team to route final CAD assembly and optimize heater and RTD sensor placement.
- Automated harness CAD routing by using **Python** scripting, traversing **Capital Essentials** wiring diagrams to streamline **Nx** CAD assembly process of harnessing with mapped ports and ID instances to optimize constraints, clocking of connectors and backshells, and generate formboard drafting (cut list, BOM, resistance measurements, etc.) efficiently.
- Create BOM and collaborated with technicians to build and send test harnesses for optical systems in Germany.

Aynak, Inc. January 2025 - August 2026
Mechanical Engineering Intern

- Spearheaded design and hardware integration of 0.11 mm thin speaker **flex PCBs**, optimizing stack-up thickness and routing through 3 mm heat-set hinges within smart-eyewear frames housing 4 development boards and 1 battery.
- Designed and **SLA** printed custom mounts for 3 distinct speaker configurations, used in experimental testing of beamformer sound transmission. Optimized mount geometries and hardware placement to minimize howling and acoustic feedback.
- Prototyped hearing aid receiver mount integrated into smart eyewear frames of sub-3 mm internal clearance.
- Performed quality assurance and testing of actuators for bone conduction hearing within rigid components of eyewear frame.

Luan Lab, University of California San Diego January 2025 - June 2026
Undergraduate Researcher https://sites.google.com/ucsd.edu/luangroup

- Implementing mechanical guided assembly into wearable and flexible electronics such as skin sensors and biofeedback mechanisms. Heading experimental procedures on morphing systems and programmable structures.
- Utilized Gallium-Indium synthesis approach for electromagnetically reconfigured channels within substrates.
- Fabricated 5 mm wide substrates via **spincoating** and 4 custom **PLA** mold patterns designed in **SolidWorks**.
- Aiming to develop novel electromagnetism-based methods for patient specific morphing healthcare devices.

MAE 3 Course Tutor March 2025 - June 2026
Design Studio Tutor

- Redesigned 1 of 2 project fabrication procedures for **Fusion 360 CAD** of pendulum clock, developing smoothed replacement **CAD** models, creating labeled GIF graphics and video tutorials, and modeling common errors with prevention strategies.
- Constructed “World cup” robot combat arena in AutoCAD, lasercut and assembled with T-Slot frames and material constraints.
- Designed a geared motor adapter and demo for ongoing lab curriculum to demonstrate press fit and linkages in lecture.
- Maintained fabrication studio, assist in instrumentation troubleshooting, and maintain **LaserCAMM** and Bambu **3D printers**.

Instructional Assistant, Lab Tutor

- Provided mentorship to a total of 19 students on **Fusion 360, AutoCAD,** and **GD&T**, leading 2 fabrication projects in studio.
- Taught manufacturing, fabrication, and **CAD** skills, including **Fusion 360**, **AutoCAD**, laser cutting, and **3D printing**.

Vi at La Jolla Village Senior Living July 2023 - April 2026
Server. Adapted to dual roles as server and hostess

RELEVANT PROJECTS

Triton Robotics January 2024 - June 2025
Portable Volleyball Flywheel Summer 2025

SKILLS

Technical Skills: GD&T, DFM, machine shop, Python (Pandas, Tkinter), SLA/FDM 3D printing, flex PCB integration
Software & Tools: Siemens Nx CAD (Electrical routing), Teamcenter, Capital Essentials, SolidWorks (CAM, motion analysis, FEA), Fusion 360 (Mesh, Surface, Sheet Metal), AutoCAD, ANSYS (FEA), MATLAB, Excel, Visual Studio Code